

## HW 5.4.1: Applying the Power Rule

Find  $f'(x)$ .

1.  $f(x) = 3x^4$

2.  $f(x) = 5x^2 - 3x + 6$

3.  $f(x) = \frac{4}{x}$

4.  $f(x) = -\frac{5}{x^2}$

5.  $f(x) = 6\sqrt{x}$

6.  $f(x) = \frac{3}{2}\sqrt[3]{x}$

7.  $f(x) = 5x\sqrt{x}$

8.  $f(x) = \sqrt{x^5} + 7$

9. A projectile is shot upward with an initial velocity of 28 ft/sec from the top of an 8 foot ladder. The equation for the height of the projectile after  $t$  seconds is  $h(t) = -16t^2 + 28t + 8$ .

a. The velocity function (or rate of change of the height) is the derivative of  $h(t)$ . Find this derivative function.

b. Is the height of the projectile increasing or decreasing after 1.5 seconds?

c. The acceleration of the projectile is the rate of change of the velocity function. What is the acceleration function for the projectile?

10. The ideal gas law from chemistry is  $PV = nRT$ , which can be rewritten as  $P(V) = \frac{nRT}{V}$  if we were interested in seeing the effects of changing the volume would have on pressure.

a. Find the rate of change of  $P$  with respect to volume.

b. Discuss why the *sign* of your answer to (a) makes sense.

Answers:

1.  $f'(x) = 12x^3$

2.  $f'(x) = 10x - 3$

3.  $f'(x) = -\frac{4}{x^2}$

4.  $f'(x) = \frac{10}{x^3}$

5.  $f'(x) = \frac{3}{\sqrt{x}}$

6.  $f'(x) = \frac{1}{2\sqrt[3]{x^2}}$

7.  $f'(x) = \frac{15\sqrt{x}}{2}$

8.  $f'(x) = \frac{5\sqrt{x^3}}{2}$

9. a.  $h'(t) = -32t + 28$

b. decreasing

c.  $a(t) = v'(t) = h''(t) = -32$

10. a.  $P'(V) = -\frac{nRT}{V^2}$

b. The pressure is inversely proportional to volume. In other words, as the volume increases the pressure decreases. The negative sign of  $P'$  means  $P$  is decreasing.